MCU

USB_DP
USB_DM
NRST
BOOT0
SWDIO
SWCLK

SDMMC_D0
SDMMC_D1
SDMMC_D2
SDMMC_D3
SDMMC_CMD
SDMMC_CK

IMU_I2C_SCL
IMU_I2C_SDA
ACCEL_EXTI
GYRO_EXTI

DIP0
DIP1
DIP2
DIP3
DIP4
DIP5

USER_EXTI0
USER_EXTI1
USER_EXTI2

LED_SPI_SCK
LED_SPI_MOSI

MOTOR_SPI_SCK
MOTOR_SPI_MISO
MOTOR_SPI_MOSI
MOTOR0_SPI_CS
MOTOR1_SPI_CS
MOTOR2_SPI_CS
MOTOR3_SPI_CS
DRIBBLER_SPI_CS
KICKER_I2C_SCL
KICKER_I2C_SDA
EXPANSION_SCL
EXPANSION_SDA
OLED_I2C_SDA
OLED_I2C_SCL
ESP_SPI_SCK
ESP_SPI_MOSI
ESP_SPI_MISO
ESP_SPI_CS
ESP_HANDSHAKE
ESP_DATA_READY
ESP_RST

PWR_UART_TX
PWR_UART_RX
PWR_SRC_5V
PWR_SRC_3V3

File: 02_MCU.kicad_sch

USB and Debug

USB_DP
USB_DM
NRST
BOOT0
SWDIO
SWCLK

File: 03_USB_Debug.kicad_sch

SD Card Storage

SDMMC_D0
SDMMC_D1
SDMMC_D2
SDMMC_D3
SDMMC_CMD
SDMMC_CK

File: 04_Storage.kicad_sch

IMU

IMU_I2C_SCL
IMU_I2C_SDA
ACCEL_EXTI
GYRO_EXTI

File: 05_IMU.kicad_sch

Robot ID DIP

DIP0
DIP1
DIP2
DIP3
DIP4
DIP5

File: 08_DIP.kicad_sch

User Buttons

USER_EXTI0
USER_EXTI1
USER_EXTI2

File: 09_Buttons.kicad_sch

DotStar LEDs

LED_SPI_SCK
LED_SPI_MOSI

File: 06_LEDs.kicad_sch

Motor, Kicker, Expansion, OLED, Radio

MOTOR_SPI_SCK
MOTOR_SPI_MISO
MOTOR_SPI_MOSI
MOTOR0_SPI_CS
MOTOR1_SPI_CS
MOTOR2_SPI_CS
MOTOR3_SPI_CS
DRIBBLER_SPI_CS
KICKER_I2C_SCL
KICKER_I2C_SDA
EXPANSION_SCL
EXPANSION_SDA
OLED_I2C_SDA
OLED_I2C_SCL
ESP_SPI_SCK
ESP_SPI_MOSI
ESP_SPI_MISO
ESP_SPI_CS
ESP_HANDSHAKE
ESP_DATA_READY
ESP_RST

File: 07_Connectors.kicad_sch

Power

PWR_UART_TX
PWR_UART_RX
PWR_SRC_5V
PWR_SRC_3V3

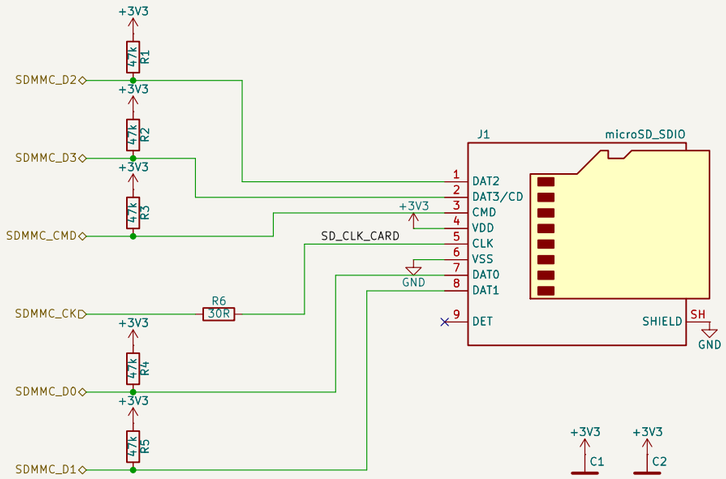
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Sheet: /
File: ControlBoard2027.kicad_sch

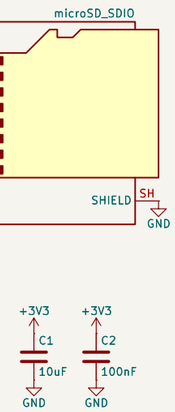
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Size: A2	Date:	**Rev:**
KiCad E.D.A. 10.0.0		Id: 1/10

MICROSD J1 ON SDMMC1 4-BIT (NO CARD DETECT)



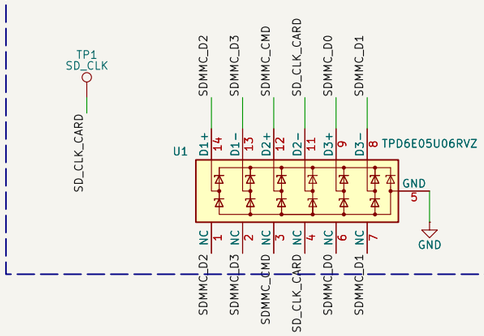
R6 = 30R provisional CLK source damping near PC12.
CMD and DAT0-3 currently have no series damping; check both directions.
47k pull-ups are bias, not impedance termination.
No data sheet establishes a constant 25R MCU output impedance.



SD CARD: select an actual card MPN before final timing approval.
Kingston SDCIT example: identification clock <=400 kHz, NOT 8 MHz.
Higher transfer rates require successful initialization and timing validation.
J1 Hirose DM3AT-SF-PEJM5. Shell to GND; DET unused; power cannot be cycled independently.
Card recovery and shell/ESD treatment remain verification items.

SDMMC LAYOUT: review CLK, CMD and DAT0-3 as one timing interface.
Routed ~50R over In1 GND; CLK/CMD/DAT0-3 matched 44.4-45.2 mm incl. R6.
Keep return paths continuous, short stubs and local J1 bypass capacitors.
Use DS13313 SDMMC timing plus the selected card's data sheet.

SD ESD U1 AT J1, CLK TEST POINT



Sheet: /SD Card Storage/
File: 04_Storage.kicad_sch

Title:

Size: A3 | Date: | **Rev:**
KiCad E.D.A. 10.0.0 | Id: 2/10

VDD DECOUPLING (EACH CAP AT ITS PIN)

ROUTING — DECOUPLING: each cap beside the STM32 pin named under it, via straight to GND.
C22/C23 (VCAP) tight to p71/p106. The netlist does not keep this pairing.

+3V3 — C4 100nF — C6 100nF — C8 100nF — C10 100nF — C12 100nF — C14 100nF — C15 100nF — C16 100nF — C17 100nF — C21 100nF — GND

VDD p17 | VDD p30 | VDD p39 | VDD p52 | VDD p62 | VDD p72 | VDD p84 | VDD p108 | VDD p121 | VDD p131

+3V3 — C3 100nF — C5 10uF 10V X5R — C7 4.7uF 16V X5R — C9 100nF — C11 4.7uF 16V X5R — C13 100nF — GND

VDD p144 | VDD bulk (C5+C7) not pin-specific | VBAT p6 | VDD33USB p95 (C11+C13) USB FS PHY — keep both close

VDDA / VREF+ DECOUPLING

VDDA p33 and VREF+ p32 use +3V3.
Separate local bypass pairs; analog noise must be measured.
Source: DS13313 supply scheme / electrical characteristics.

TP2 TP_VDDA — GND

C24 1uF 10V X5R — C25 100nF — C26 1uF 10V X5R — C27 100nF

VDDA p33 (C24+C25) | VREF+ p32 (C26+C27)

VBUS SENSE

VBUS — R20 10K — VBUS_SENSE — R21 10K — GND

HSE 25MHZ CRYSTAL

ROUTING — HSE CRYSTAL
X1, C18, C19 within 5mm of PH0/PH1.
Short equal OSC traces, GND guard ring.
Nothing routed under or beside the loop.

X1 25MHz 10pF — C18 12pF — C19 12pF — R12 0R

CLOCK: X1 = 25 MHz: set firmware clock tree to the fitted part.
R12 = 0R is an oscillator drive-tuning link, not line termination.
C18/C19 need measured stray capacitance to establish the 10 pF load.
Startup margin and crystal drive remain unverified (ABM8, DS13313 HSE data).
No blanket claim that 25 MHz always starts or HSI48 is unusable.

SUPPLIES: PDR_ON and VBAT are on +3V3; no backup battery.
VCAP 71/106 share their bypass node: retained pending ERC symbol review.
Do not separate a required supply node merely to silence ERC.
Source: DS13313 power-supply scheme / VCAP data.

U2 — STM32H723ZGT6

NRST — C20 100nF — GND
BOOT0 — R11 10K — R5 — GND
VREF+ — +3V3
PDR_ON — +3V3

R15/R16 = 30R: provisional source damping, not certified matching.
No fixed driver resistance or 55R target follows from DS13313 Tables 52/54.
Validate SCK/MOSI/MISO/CS at every cable receiver, both directions.

ROUTING — MOTOR SPI (6 Mbit/s, 5 drops)
R15/R16 within 5mm of PA5/PA7, not at the connectors.

PA0 PWR_UART_TX
PA1 PWR_UART_RX
PA2 PWR_SRC_5V
PA3 PWR_SRC_3V3
PA5 — R15 30R — MOTOR_SPI_SCK
PA6 MOTOR_SPI_MISO
PA7 — R16 30R — MOTOR_SPI_MOSI
PA9 VBUS_SENSE
PA11 USB_DM
PA12 USB_DP
PA13_SWDIO SWDIO
PA14_SWCLK SWCLK — R13 10K — +3V3
PA15_JTDI
PB0 MOTOR1_SPI_CS
PB1 MOTOR2_SPI_CS
PB2 MOTOR3_SPI_CS
PB3_SWO — R65 30R — ESP_SPI_SCK
PB4_NJTRST ESP_SPI_MISO
PB5 ESP_SPI_CS
PB6 ESP_HANDSHAKE
PB7 ESP_DATA_READY
PB8 OLED_I2C_SCL
PB9 OLED_I2C_SDA
PB10 KICKER_I2C_SCL
PB11 KICKER_I2C_SDA
PB13 LED_SPI_SCK
PB15 LED_SPI_MOSI
PC0 DIP4
PC1 DIP5
PC4
PC5 MOTOR0_SPI_CS
PC6
PC7
PC8 SDMMC_D0
PC9 SDMMC_D1
PC10 SDMMC_D2
PC11 SDMMC_D3
PC12 SDMMC_CK
PC13 USER_EXTI1
PC14 USER_EXTI2
PD0
PD1
PD2 SDMMC_CMD
PD3
PD4
PD5
PD6 — R66 30R — ESP_SPI_MOSI
PD7
PD8
PD9
PD10
PD11
PD12
PD13
PD14
PD15

+3V3 pull-ups: R7 2.2k, R8 2.2k, R9 2.2k, R10 2.2k, R14 2.2k, R17 2.2k, R18 2.2k, R19 2.2k

I2C: R7/R9, R8/R10, R14/R17, R18/R19 are each 2.2k.
At 3.3 V: (3.3−0.4)/2.24 = 1.32 mA sink requirement.
For a 300 ns rise-time limit: C <= 300ns/(0.8473R) = 161 pF nominal.
At +5% R: 153 pF. Include ESD, pins, cables and parallel pull-ups.
LSM6DSV320X / SSD1306 / DS13313 timing and I/O data: external loads not qualified.

I2C VALIDATION REQUIRED: measure SDA and SCL at worst load.
Do not change to 1.5k from rise time alone: recheck every device's VOL/current.
OLED module pull-ups and expansion-device specifications are not established.

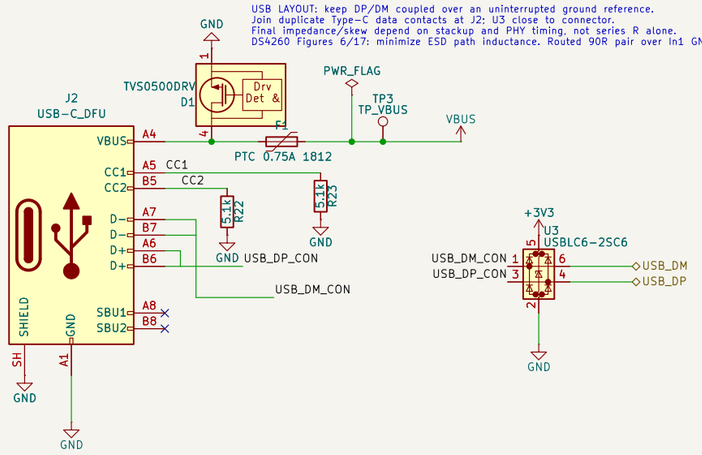
IMU_I2C_SDA0 — R7 2.2k
IMU_I2C_SCL0 — R9 2.2k
GYRO_EXTI0
ACCEL_EXTI0
DIP0 DIP1 DIP2 DIP3
DRIBBLER_SPI_CS — R10 2.2k
EXPANSION_SCL0 — R8 2.2k
EXPANSION_SDA0
ESP_RST
USER_EXTI0

VCAP — C22 2.2uF — GND; C23 2.2uF — GND

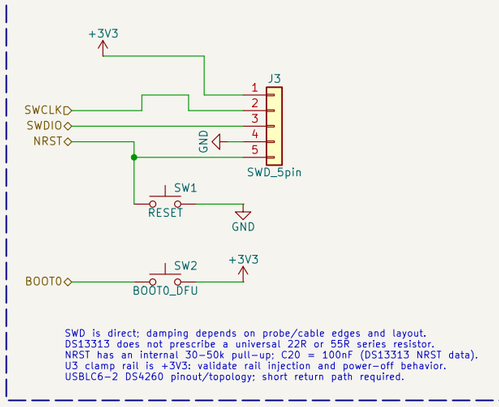
VSS VSSA — 16 31 — GNDGND

Sheet: /MCU/	
File: 02_MCU.kicad_sch	
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KiCad E.D.A. 10.0.0	Rev: Id: 3/10

USB-C J2: TVS, FUSE, CC RESISTORS, ESD



SWD DEBUG J3, RESET AND BOOT BUTTONS



USB LOAD: F1 PTC Ihold = 0.75 A at 20–23C, 0.55 A at 50C.
Actual concurrent worst-case board/external current is not established.
Restricted bench use until host current, startup and thermal budget are checked.
F1 does not negotiate USB current; replacing it alone cannot approve more load.

Sheet: /USB and Debug/
File: 03_USB_Debug.kicad_sch

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Size: A3	Date:	**Rev:**
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MOTOR MODULES J4–J8

J4
GND 1
MOTOR_SPI_SCK 2
GND 3
MOTOR_SPI_MOSI 4
MOTOR_SPI_MISO 5
MOTOR0_SPI_CS 6
MOTOR0_WHEEL

J5
GND 1
MOTOR_SPI_SCK 2
GND 3
MOTOR_SPI_MOSI 4
MOTOR_SPI_MISO 5
MOTOR1_SPI_CS 6
MOTOR1_WHEEL

J6
GND 1
MOTOR_SPI_SCK 2
GND 3
MOTOR_SPI_MOSI 4
MOTOR_SPI_MISO 5
MOTOR2_SPI_CS 6
MOTOR2_WHEEL

J7
GND 1
MOTOR_SPI_SCK 2
GND 3
MOTOR_SPI_MOSI 4
MOTOR_SPI_MISO 5
MOTOR3_SPI_CS 6
MOTOR3_WHEEL

J8
GND 1
MOTOR_SPI_SCK 2
GND 3
MOTOR_SPI_MOSI 4
MOTOR_SPI_MISO 5
DRIBBLER_SPI_CS 6
DRIBBLER

I2C CONNECTORS: J10 KICKER (SIGNALS ONLY), J11/J12 FED BY F3/F4

+5V → F3 PTC 0.75A 1812 → +5V_EXT1
+5V → F4 PTC 0.75A 1812 → +5V_EXT2

J10 KICKER_I2C
1 KICKER_I2C_SDA
2 KICKER_I2C_SCL
3 GND

J11 EXPANSION_I2C_1
1 +5V_EXT1
2 EXPANSION_SCL
3 EXPANSION_SDA
4 GND

J12 EXPANSION_I2C_2
1 +5V_EXT2
2 EXPANSION_SCL
3 EXPANSION_SDA
4 GND

OLED J13

J13 SK1 Socket 1x8
1 OLED_I2C_SDA
2 OLED_I2C_SCL
3 GND
4 +3V3
5
6
7
8 GND
OLED_Adafruit_326

RADIO LINK J9 (SAMTEC LSHM MEZZANINE)

J9 RADIO_MEZZ
1 +3V3
2 ESP_RST
3
4 ESP_DATA_READY
5 GND
6 ESP_HANDSHAKE
7
8 ESP_SPI_MISO
9
10
11 ESP_SPI_SCK
12
13
14
15 ESP_SPI_MOSI
16
17
18
19 ESP_SPI_CS
20 GND
VH GND

C28 +3V3 22uF GND
C29 +3V3 100nF GND

J9 = RadioBoard2027 J2. Samtec LSHM-110-02.5 on both boards (5mm stack).
Even row: 2 RST, 4 DATA_READY, 6 HANDSHAKE, 8 MISO, 12 SCK, 16 MOSI, 20 CS.
10/14/18 GND. Odd row: 1/3/5 +3V3. 7–19 GND. SH to GND.
Face to face, pin p here mates radio pin p+1 (odd p) or p−1 (even p).
The radio hangs off the east edge; nothing holds it but the connector.
R31 100k / R32 10k hold HANDSHAKE/DATA_READY low with the radio absent or in reset.
Reset is open-drain into the radio's 10k/1uF EN network (DESIGN_NOTES §11).

CS PULL-UPS / PULL-DOWNS

MOTOR0_SPI_CS — R24 10k — +3V3
MOTOR1_SPI_CS — R25 10k — +3V3
MOTOR2_SPI_CS — R26 10k — +3V3
MOTOR3_SPI_CS — R27 10k — +3V3
DRIBBLER_SPI_CS — R28 10k — +3V3
ESP_SPI_CS — R29 10k — +3V3
MOTOR_SPI_SCK — R30 100k — GND
ESP_HANDSHAKE — R31 100k — GND
ESP_DATA_READY — R32 10k — GND

ESD AT CONNECTORS U4–U6, U12–U14

U4 TPD4E05U06DQA
NC 10
D1+ 1 ← MOTOR_SPI_SCK
D1− 2 ← MOTOR_SPI_MOSI
NC 9
GND 3
D2+ 4 → MOTOR_SPI_MISO
D2− 5 → MOTOR0_SPI_CS
NC 7
NC 6
GND

U5 TPD4E05U06DQA
MOTOR_SPI_SCK
MOTOR_SPI_MOSI
MOTOR_SPI_MISO
MOTOR1_SPI_CS
GND

U6 TPD4E05U06DQA
EXPANSION_SCL
EXPANSION_SDA
KICKER_I2C_SCL
KICKER_I2C_SDA
GND

U12 TPD4E05U06DQA
MOTOR_SPI_SCK
MOTOR_SPI_MOSI
MOTOR_SPI_MISO
MOTOR2_SPI_CS
GND

U13 TPD4E05U06DQA
MOTOR_SPI_SCK
MOTOR_SPI_MOSI
MOTOR_SPI_MISO
MOTOR3_SPI_CS
GND

U14 TPD4E05U06DQA
MOTOR_SPI_SCK
MOTOR_SPI_MOSI
MOTOR_SPI_MISO
DRIBBLER_SPI_CS
GND

SCOPE TEST POINTS

TP4 SCK — MOTOR_SPI_SCK
TP5 MOSI — MOTOR_SPI_MOSI
TP6 MISO — MOTOR_SPI_MISO
TP7 KICK_SCL — KICKER_I2C_SCL
TP8 EXP_SCL — EXPANSION_SCL
TP9 GND — GND

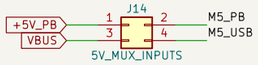
OLED MOUNTING HOLES H1–H4

H1 MountingHole
H2 MountingHole
H3 MountingHole
H4 MountingHole

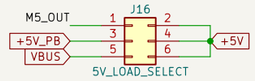
SPI ROUTING: routed -50R over In1 GND: receiver data still pending.
R15/R16 are provisional damping; verify all multidrop endpoints.
Place ESD devices at connectors with short ground returns.
F3 -> J11, F4 -> J12: independent PTC output rails. J10 carries no power:
the kicker board is supplied by the powerboard.
0.75 A PTCs: each holds 0.55 A at 50C; NOT a combined available budget.
Shared UB/source limits still apply: fault selectivity is not guaranteed.
Shared I2C signals remain vulnerable to any device holding a bus line low.

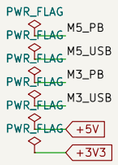
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File: 07_Connectors.kicad_sch
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Size: A2 | Date:
KiCad E.D.A. 10.0.0 | Rev: Id: 5/10

5V POWER PATHS: J14 INPUT LINKS / J16 LOAD SELECT



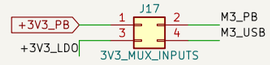
J14/J17: fit BOTH caps for normal operation.
Remove ALL FOUR input caps for failed-mux recovery.

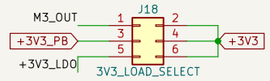




SH1	SH2	SH3	SH4	SH5	SH6
Shunt	Shunt	Shunt	Shunt	Shunt	Shunt

3V3 POWER PATHS: J17 INPUT LINKS / J18 LOAD SELECT



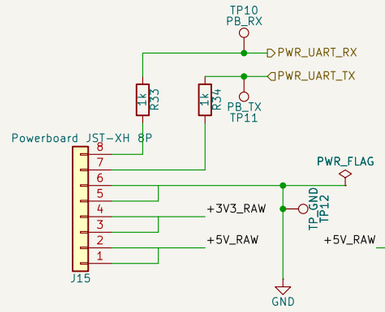


J16/J18: ONE cap each, same position on both.
1-2 MUX | 3-4 POWERBOARD | 5-6 USB. No cap = OFF.
POWER OFF before moving caps. Multiple caps SHORT SOURCES.

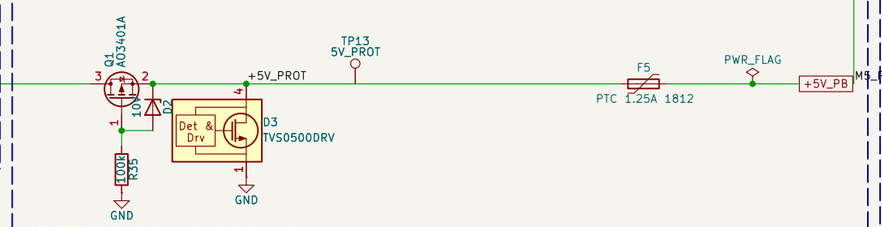
PWR_FLAGs after jumpers assume the documented caps are installed.
They are ERC source assertions, not permanent electrical bridges.

POWER LAYOUT: PCB routed; thermal verification pending.
Minimize TVS return inductance; input/output caps close to U8/U9.
Size copper from actual load, temperature and stackup, not a fixed 1 mm rule.
TPS2116 section 10; AP7361C thermal data; TVS0500 layout guidance.

POWERBOARD CONNECTOR J15

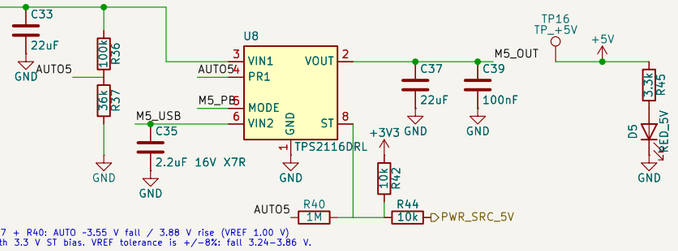


5V INPUT: Q1 REVERSE POLARITY, D3 TVS, F5 FUSE



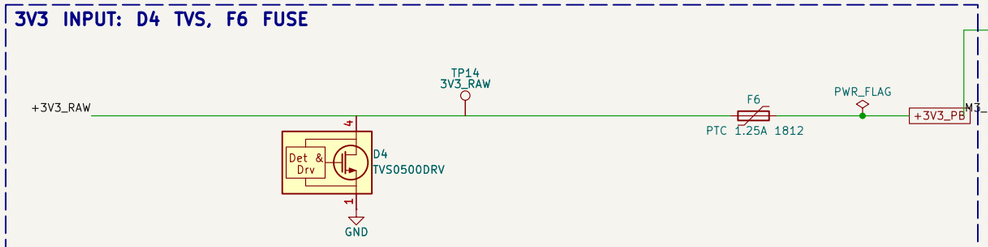
D2: BZX384-C10 gate clamp: characterize transients.
10 V rating is tested at 5 mA, not the R35 bias current.

5V MUX U8: AUTO; BYPASS AT J16

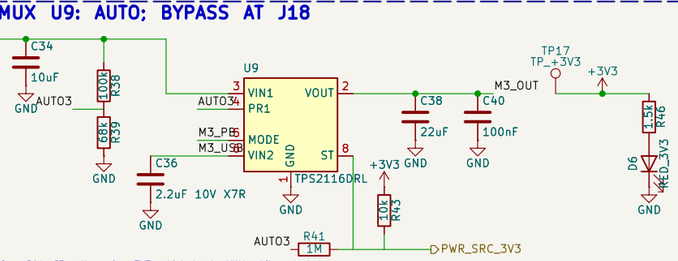


R36/R37 + R40: AUTO ~3.55 V fall / 3.88 V rise (VREF 1.00 V)
only with 3.3 V ST bias. VREF tolerance is +/-8%: fall 3.24-3.86 V.

3V3 INPUT: D4 TVS, F6 FUSE

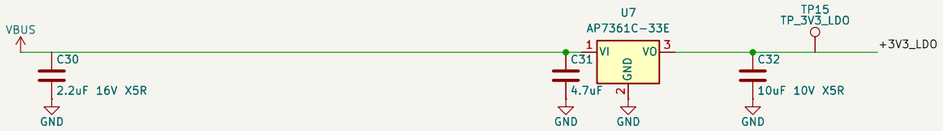


3V3 MUX U9: AUTO; BYPASS AT J18



R38/R39 + R41: ST pull-up is +3V3, which tracks VIN1 while selected.
Nominal fall ~2.34 V, rise ~2.57 V; NOT rail-valid detection.

USB VBUS AND 3V3 LDO U7



USB INPUT: C30+C31 = 6.9uF on VBUS; C35 adds 2.2uF via J14 (9.1uF < 10uF USB limit).
Bypass has no mux soft-start or reverse blocking; validate inrush and load.
PTC rating is not USB-current permission. USB 3V3 still uses U7 LDO.
Sources: TPS2116 7.3.2; MF-MSMF / 1812L; AP7361C.

PROTECTION LIMITS: TVS0500 clamps up to 9.2 V at 43 A.
This is NOT proof of protection for 6 V-absolute-max TPS2116 or 3V3 loads.
PTCs are thermal devices, not fast current limits; verify fault coordination.
Sources: TVS0500, TPS2116 absolute ratings, MF-MSMF / 1812L.

PHYSICAL BYPASS: J16/J18 carry load current, not PR1 control.
For recovery: power OFF, remove all J14/J17 caps, select PB or USB on J16/J18.
Bypass omits mux reverse blocking / soft-start. Source fuses remain.
ST reports MUX state only; cannot identify bypass source or USB current budget.
Control/status resistors remain: NOT galvanic or arbitrary-fault isolation.
Sources: TPS2116 SLVSFG1A 7.3 / 7.6.1; Samtec TSW and SNT datasheets.

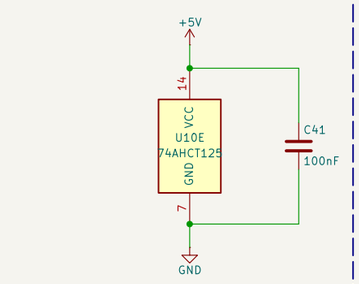
J15: 1-2 +5V; 3-4 +3V3; 5-6 GND; 7 TX; 8 RX.
Mate/cable pinout and load remain external-interface requirements.
Q1 polarity agrees with AO3401A pinout. No 3V3 reverse-polarity FET.
R33/R34 limit current; they do not guarantee powered-off isolation.
PTC faults and mux source change can still interrupt other loads.

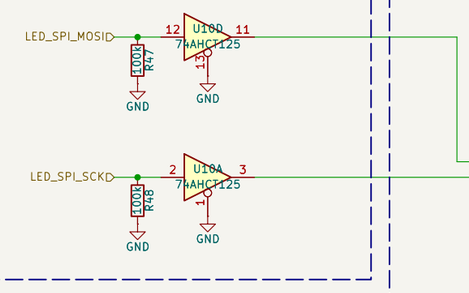
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File: 01_Power.kicad_sch

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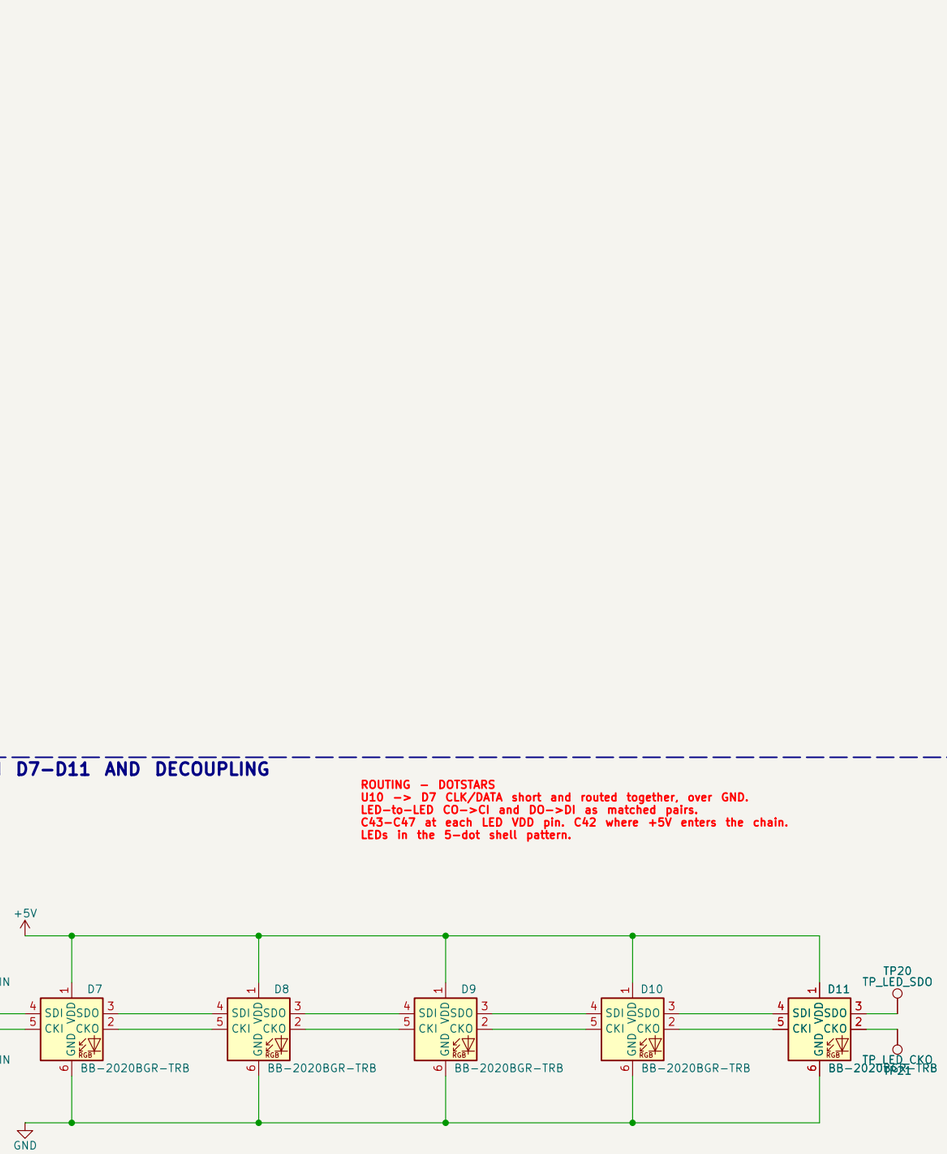
LEVEL SHIFT U10 (3.3V SPI TO 5V)

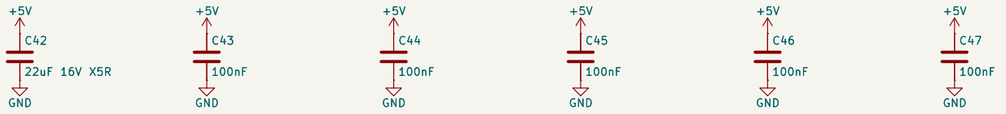




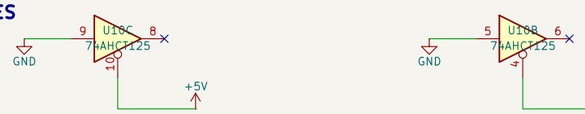
DOTSTAR CHAIN D7–D11 AND DECOUPLING

ROUTING – DOTSTARS
U10 -> D7 CLK/DATA short and routed together, over GND.
LED-to-LED CO->CI and DO->DI as matched pairs.
C43–C47 at each LED VDD pin. C42 where +5V enters the chain.
LEDs in the 5-dot shell pattern.





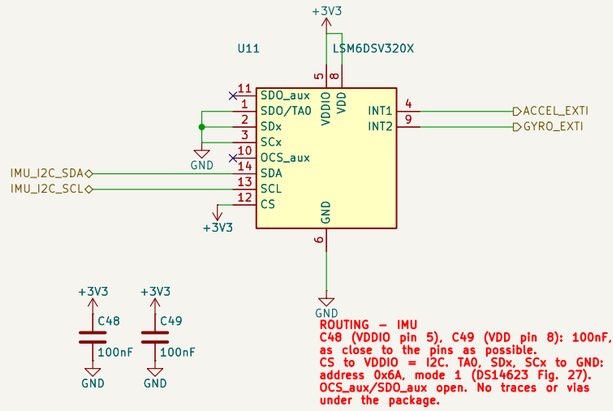
U10 SPARE GATES



LED LEVEL SHIFT: U10 = SN74AHCT125, specified for VCC 4.5–5.5 V.
3.3 V logic satisfies TTL input levels at a valid U10 supply.
R47/R48 = 100k bias inputs low; unused gates disabled.
D7–D11 = American Bright BB-2020BGR-TRB; use its data sheet/pad drawing.
A muxed USB source does NOT guarantee >=4.5 V after all cable/fuse drops.
Firmware current caps are requirements, not evidence of implemented limits.
Verify LED startup/current, power budget and custom footprints.
TP18/19 check chain input; TP20/21 check chain output.

Sheet: /DotStar LEDs/	
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IMU U11 LSM6DSV320X: I2C (0x6A), INT1/INT2 INTERRUPTS



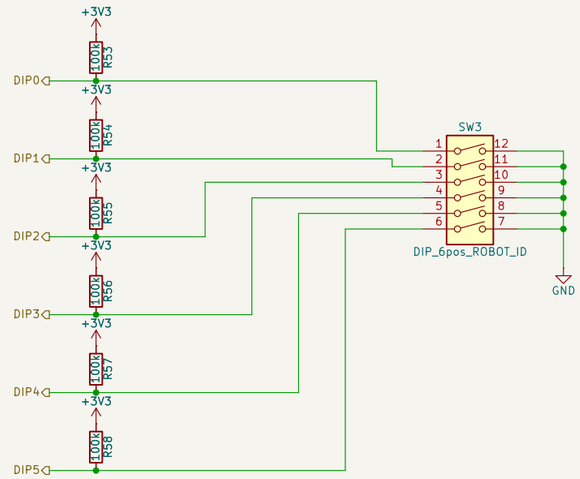
INTERRUPTS: INT1 -> ACCEL_EXTI (PF3). INT2 -> GYRO_EXTI (PF2). Both are push-pull and are driven low from power-up (DS14623 Table 25), so no pull-downs are fitted.
Route data-ready or events to them with INT1_CTRL/INT2_CTRL.

Sheet: /IMU/
File: 05_IMU.kicad_sch

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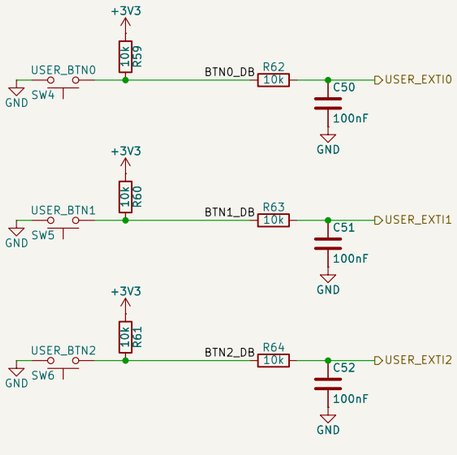
ROBOT ID DIP SWITCH SW3 (100K PULL-UPS)



DIP R53–R58 = 100k: STM32 250 nA input leakage gives 25 mV drop nominal.
Logic bias is adequate for an unloaded MCU input (DS13313 Table 51).
DIP switch: verify minimum contact load: 33 uA is not proof.
Debounce/sample stability needs firmware; no universal 100nF requirement.

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USER BUTTONS SW4–SW6 WITH RC DEBOUNCE



BUTTONS: R59–R61 pull up with 10k; R62–R64 series 10k; C50–C52 = 100nF.
Calculated nominal time constants: press 1 ms, release 2 ms.
At +/−1% R and +/−10% C: 0.891–1.111 ms and 1.782–2.222 ms.
BOM: R59–R64 are 1%; C50–C52 need a +/−10% X7R part.
RC filtering does not guarantee removal of all bounce; qualify switch and firmware.
Series R is between switch node and MCU/cap node, not between cap and MCU.
Input leakage check uses STM32H723 DS13313 Table 51.

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